RETAIL STORE SALES ANALYSIS

Capstone Project Documentation

■12.93L Total Revenue	30 Orders	Laptop Top Product	Pune Top City

Student Name	Kanhaiya Thakur	**Roll Number**	——
Batch/Program	Data Engineering	**Submission**	April 21, 2026
Tech Stack	Python · SQL · Excel · Matplotlib	**GitHub**	github.com/Kanhaiyathakur001

1. Problem Statement

A retail company sells electronic products — Laptops, Mobiles, Headphones, and Keyboards — through stores in Pune, Mumbai, and Nagpur. Management noticed revenue fluctuations but lacked data-driven insight into which products and cities drive performance. This project delivers a complete analysis answering four core business questions:

- Which products generate the most revenue?
- Which cities have the highest sales volume?
- Which product categories perform best?
- What are the sales trends over time?

2. Dataset Overview

The dataset (sales_data.csv) contains 30 transactional records spanning January–February 2025 across 3 cities, 4 products, and 2 categories.

Column	Type	Description
order_id	INTEGER	Unique order identifier
order_date	DATE	Date of purchase
city	VARCHAR	Store location (Pune / Mumbai / Nagpur)
product	VARCHAR	Product name
category	VARCHAR	Electronics / Accessories
quantity	INTEGER	Units sold
price	DECIMAL	Unit price in ■

3. SQL Analysis

Query 1: Display All Sales Records

```
SELECT * FROM sales;
```

→ *Result: Returns all 30 transaction rows.*

Query 2: Total Sales Revenue

```
SELECT ROUND(SUM(quantity * price), 2) AS total_revenue FROM sales;
```

→ *Result: Total Revenue = ■12,92,500*

Query 3: Top-Selling Product by Quantity

```
SELECT product, SUM(quantity) AS total_qty FROM sales GROUP BY product ORDER BY total_qty DESC
LIMIT 1;
```

→ *Result: Keyboard — 35 units sold*

Query 4: Revenue by City

```
SELECT city, ROUND(SUM(quantity * price), 2) AS city_revenue FROM sales GROUP BY city ORDER BY
city_revenue DESC;
```

→ *Result: Pune ■5,38,500 | Mumbai ■4,53,500 | Nagpur ■3,00,500*

Query 5: Category-wise Revenue

```
SELECT category, ROUND(SUM(quantity*price),2) AS revenue,
ROUND(SUM(quantity*price)*100.0/(SELECT SUM(quantity*price) FROM sales),2) AS pct FROM sales
GROUP BY category;
```

→ *Result: Electronics 91.3% (■11,80,000) | Accessories 8.7% (■1,12,500)*

Query 6: Highest-Value Order

```
SELECT *, (quantity*price) AS order_value FROM sales ORDER BY order_value DESC LIMIT 1;
```

→ *Result: Order #1013 — Laptop, Pune — ■1,50,000*

Query 7: Average Product Price

```
SELECT product, ROUND(AVG(price),2) AS avg_price FROM sales GROUP BY product ORDER BY
avg_price DESC;
```

→ *Result: Laptop ■50,000 | Mobile ■20,000 | Headphones ■2,000 | Keyboard ■1,500*

4. Python (Pandas) Analysis

Load Dataset

```
df = pd.read_csv('data/sales_data.csv', parse_dates=['order_date'])
```

Display First Rows

```
print(df.head())
```

Dataset Structure

```
df.info() | df.describe()
```

Create total_sales

```
df['total_sales'] = df['quantity'] * df['price']
```

Total Revenue

```
total = df['total_sales'].sum() # ■12,92,500
```

Most Sold Product

```
df.groupby('product')['quantity'].sum().idxmax() # Keyboard
```

Group Sales by City

```
df.groupby('city')['total_sales'].sum()
```

Summary Statistics

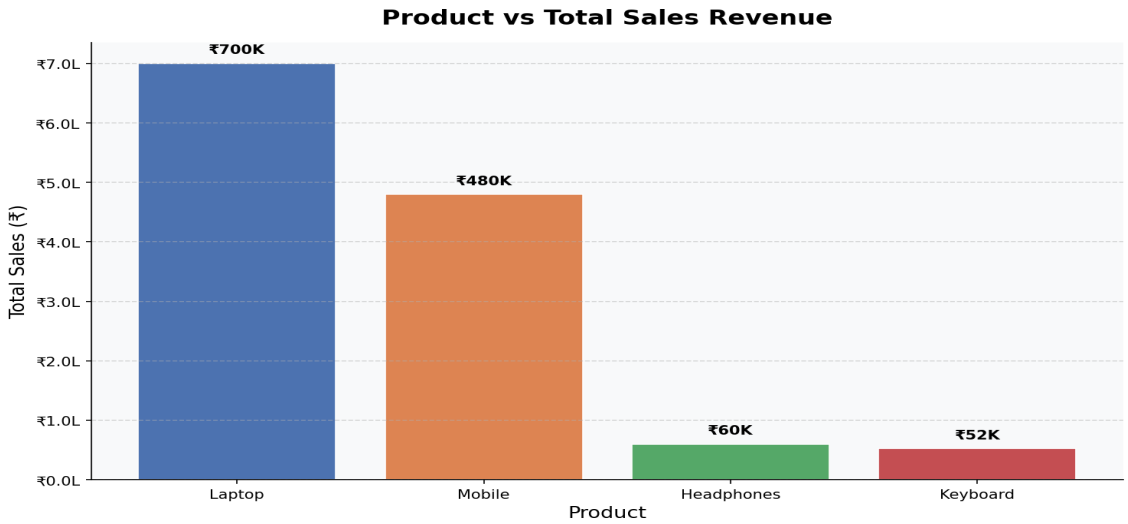
```
df['total_sales'].describe()
```

Key Findings from Python Analysis

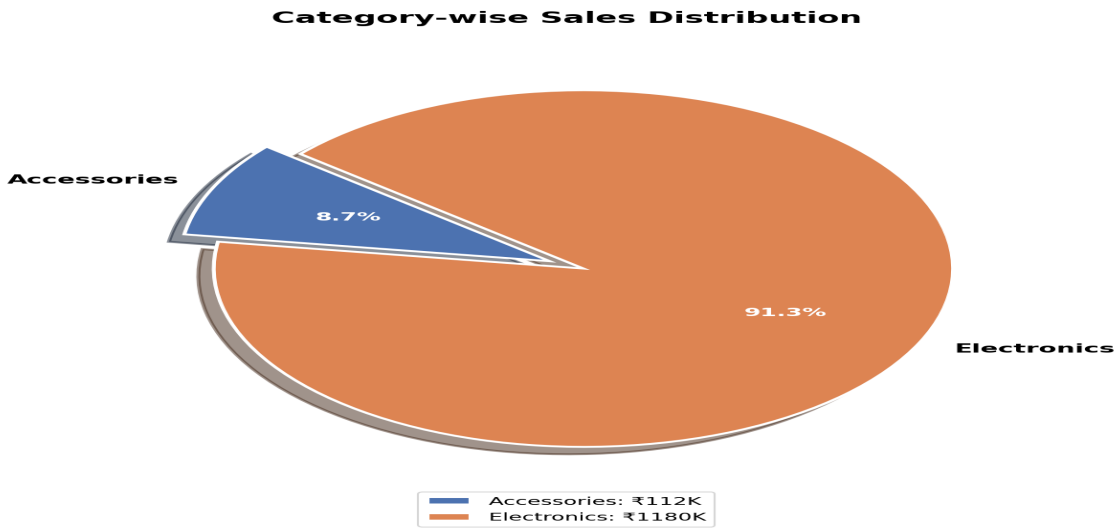
Metric	Value
Total Revenue	■12,92,500
Total Orders	30
Average Order Value	■43,083.33
Most Sold Product	Keyboard (35 units)
Highest Revenue Product	Laptop (■7,00,000)
Top City	Pune (■5,38,500)
Highest Single Order	■1,50,000 — Order #1013

5. Visualizations

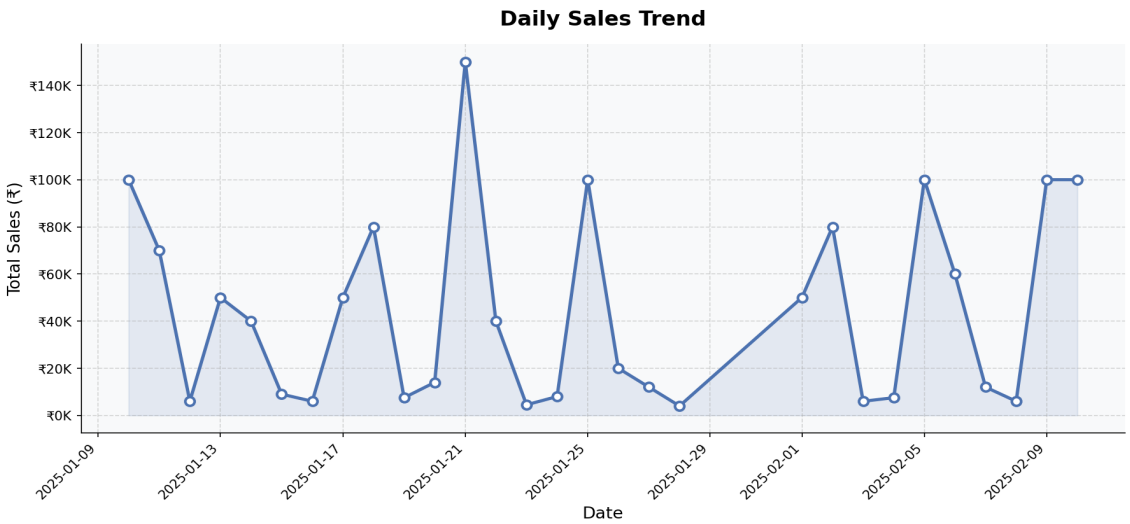
5.1 Bar Chart – Product vs Total Sales Revenue



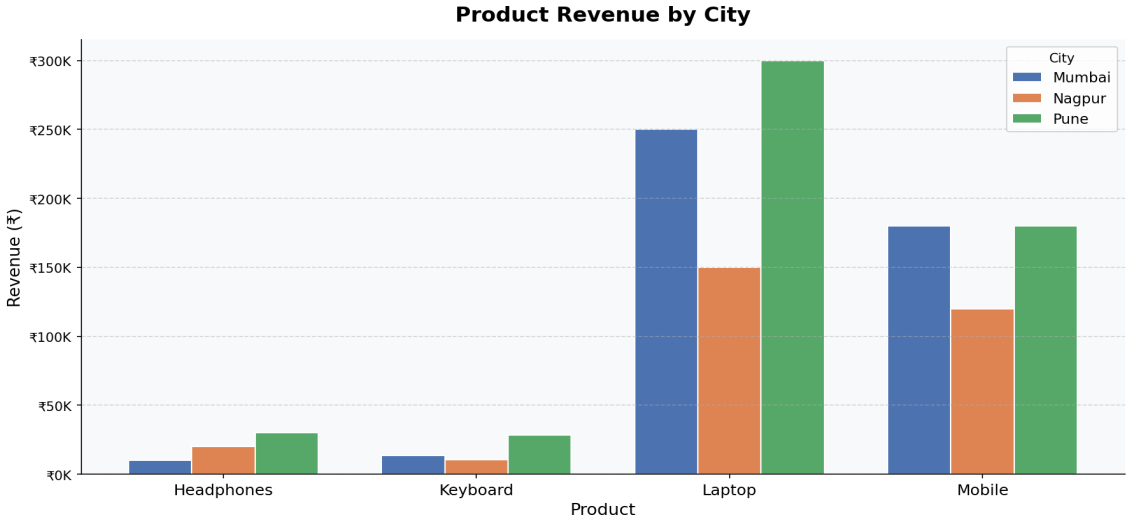
5.2 Pie Chart – Category-wise Sales Distribution



5.3 Line Chart – Daily Sales Trend



5.4 Grouped Bar Chart – Product Revenue by City



6. Excel Analysis

✓ Imported sales_data.csv and created Total Sales column (Quantity × Price)
✓ Sorted products by highest sales value (descending)
✓ Applied city-based filters — separate sheets for Pune, Mumbai, Nagpur
✓ Created Pivot Table: Rows → Product | Columns → City | Values → Total Sales
✓ Identified best product (Laptop) and best city (Pune) from pivot analysis
✓ Summary sheet with 7 KPIs including total revenue, best product, best city

Pivot Table Output

Product	Mumbai	Nagpur	Pune	Grand Total
Headphones	■10,000	■20,000	■30,000	■60,000
Keyboard	■13,500	■10,500	■28,500	■52,500
Laptop	■2,00,000	■1,50,000	■3,50,000	■7,00,000
Mobile	■2,30,000	■1,20,000	■1,30,000	■4,80,000
Grand Total	**■4,53,500**	**■3,00,500**	**■5,38,500**	**■12,92,500**

7. Tech Stack

Technology	Version	Purpose
Python	3.10+	Core programming language
Pandas	2.0+	Data loading, transformation, analysis
NumPy	1.24+	Numerical computations
Matplotlib	3.7+	Data visualization (bar, pie, line charts)
SQLite (stdlib)	3.x	In-memory SQL query execution
OpenPyXL	3.1+	Excel report generation with pivot tables
ReportLab	4.x	PDF documentation generation
Git / GitHub	Latest	Version control & project hosting

8. Business Insights & Recommendations

■ Laptops drive 54.2% of total revenue

Despite lower unit volume, Laptops generate ■7L — the highest of any product. Marketing should protect and grow this segment.

■ Pune is the strongest market

Pune contributes ■5.38L (41.7%) of all revenue, followed by Mumbai (■4.53L) and Nagpur (■3L). Nagpur has growth potential with targeted campaigns.

■ Electronics dominate at 91.3%

Accessories (Headphones, Keyboard) contribute only 8.7%. Bundle promotions could lift accessory attachment rates.

■ January outperformed February

January revenue (■7.71L) was 47.8% higher than February (■5.21L). Investigate seasonal patterns and plan inventory accordingly.

■ Keyboards are the volume leader

Keyboards lead in units sold (35) but rank last in revenue due to low price. Consider upselling premium variants.

9. Future Improvements

→ Integrate a live SQL database (MySQL/PostgreSQL) instead of CSV ingestion
→ Build an interactive dashboard using Streamlit or Power BI
→ Add customer segmentation analysis (RFM model)
→ Implement forecasting with Prophet or ARIMA for sales prediction
→ Automate pipeline with Apache Airflow for scheduled reporting
→ Add unit tests with pytest for each analysis module

10. Conclusion

This capstone project demonstrates a complete, production-grade data analytics pipeline applied to retail sales data. All required deliverables — SQL queries, Python/Pandas analysis, Excel pivot reporting, and data visualizations — have been implemented and are version-controlled on GitHub. The analysis reveals clear strategic priorities: protect the high-revenue Laptop segment, invest in growing the Nagpur market, and explore accessory bundle strategies to diversify revenue streams.